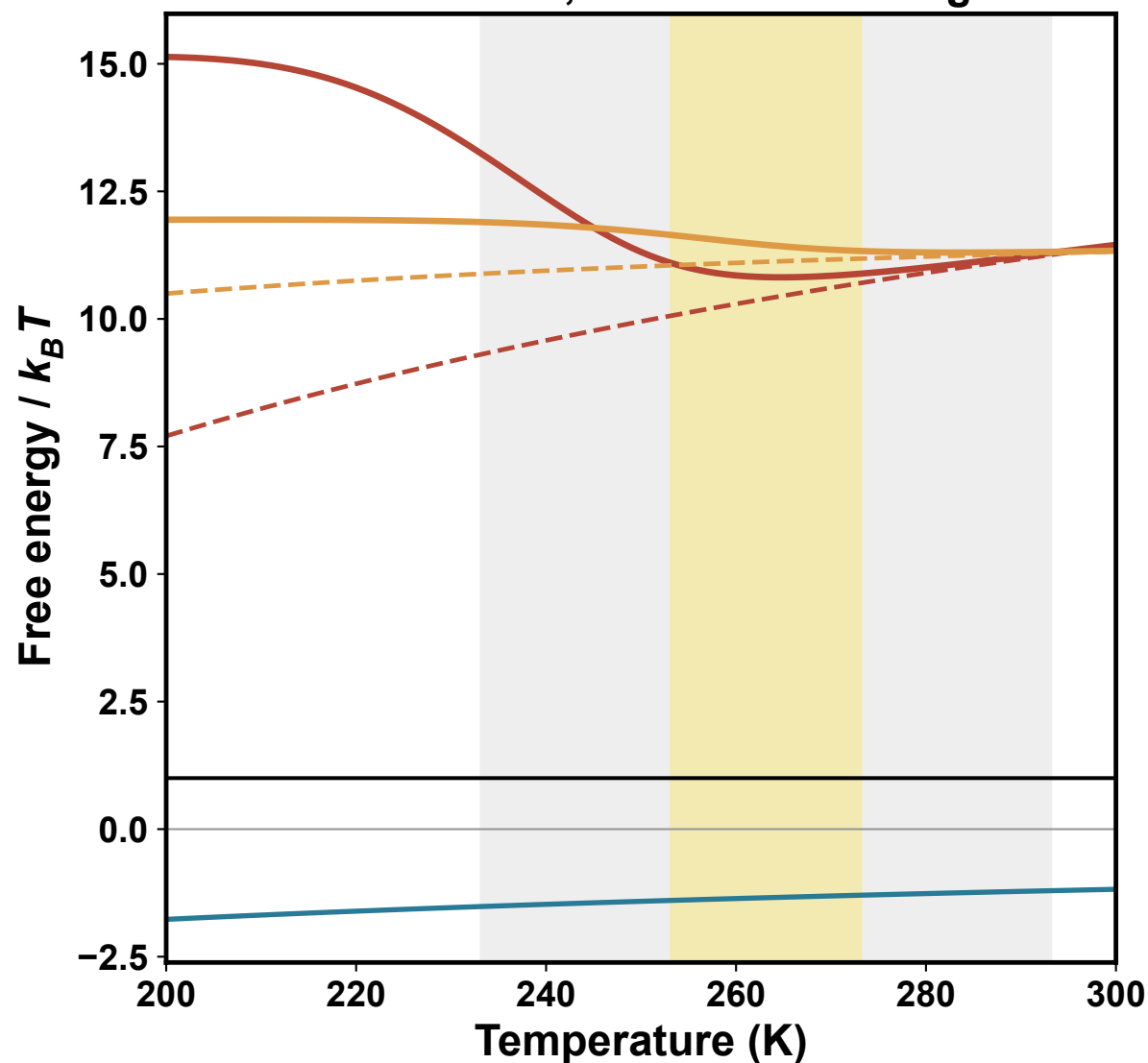


**Partial contact free energy, including vdW
20 s window, face twist +/- 5 deg**



- face, 20 s
- - - face, equilibrium
- tip, 20 s
- - - tip, equilibrium
- face vdW (component)
- $k_B T$ (reference)
- Experimental range
- Transient aggregation (experiment)